

Practical 6: Adaboost Ensemble Learning

Code:

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import matplotlib.pyplot as plt

from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.tree import DecisionTreeClassifier

from sklearn.ensemble import AdaBoostClassifier

from sklearn.metrics import accuracy_score, classification_report

# 1. Load Iris dataset

iris = load_iris()

X = iris.data

y = iris.target

# 2. Split dataset into training and testing data

X_train, X_test, y_train, y_test = train_test_split(

    X,

    y,

    test_size=0.2,

    random_state=42,

    stratify=y)

# 3. Create weak classifier

weak_classifier = DecisionTreeClassifier( max_depth=1,

    random_state=42)

# 4. Train individual weak classifier

weak_classifier.fit(X_train, y_train)

# 5. Predict using weak classifier

y_pred_weak = weak_classifier.predict(X_test)

# 6. Calculate weak classifier accuracy

weak_accuracy = accuracy_score(y_test, y_pred_weak)

print("Individual Weak Classifier Accuracy:",
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    weak_accuracy)

# 7. Create AdaBoost ensemble
adaboost_model = AdaBoostClassifier(
    estimator=DecisionTreeClassifier(
        max_depth=1,
        random_state=42 ),
    n_estimators=50,
    learning_rate=1.0,
    random_state=42)

# 8. Train AdaBoost model
adaboost_model.fit(X_train, y_train)

# 9. Make predictions
y_pred_boost = adaboost_model.predict(X_test)

# 10. Calculate AdaBoost accuracy
boost_accuracy = accuracy_score( y_test,
    y_pred_boost)

print("AdaBoost Accuracy:", boost_accuracy)

# 11. Classification report
print("\nAdaBoost Classification Report:")

print( classification_report( y_test,y_pred_boost, target_names=iris.target_names ))

# 12. Compare results
print("\nPerformance Comparison:")

print("Weak Classifier Accuracy:",weak_accuracy)

print("AdaBoost Accuracy:", boost_accuracy)

# 13. Visualization
models = ["Weak Classifier", "AdaBoost"]
accuracies = [weak_accuracy, boost_accuracy]

plt.bar(models, accuracies)

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plt.ylabel("Accuracy")

plt.xlabel("Model")

plt.title("Weak Classifier vs AdaBoost")

plt.ylim(0, 1.1)

plt.show()
```

output:

```
===== RESTART: C:/VS CODE/Python/AI/DT/AEL.py =
Individual Weak Classifier Accuracy: 0.6666666666666666
AdaBoost Accuracy: 0.9333333333333333

AdaBoost Classification Report:
              precision    recall  f1-score   support

   setosa         1.00        1.00        1.00         10
  versicolor     0.90        0.90        0.90         10
   virginica     0.90        0.90        0.90         10

 accuracy         0.93        0.93        0.93         30
 macro avg        0.93        0.93        0.93         30
weighted avg        0.93        0.93        0.93         30

Performance Comparison:
Weak Classifier Accuracy: 0.6666666666666666
AdaBoost Accuracy: 0.9333333333333333
|
```

